SEIF JABER

From a paper to a research niche

Neffke, Otto & Weyh (2017) → Joyez & Laffineur (2022) + Elekes, Tóth & Eriksson (2024) → My niche: AI & job mobility

1 THE NICHE OF NEFFKE, OTTO & WEYH (2017)

Workers move between industries all the time, but until this paper nobody had looked closely at *which* industries they actually move between. Labor economists took the high switching rates as a sign that skills aren't tied to industries, but that only works if you assume every industry is equally far from every other in the skills it needs. Older relatedness measures don't settle this, because co-location, co-exports and input-output links show where industries sit, what countries export together or who buys from whom, not what skills industries share, and the labor-flow measures people had started using had never been properly tested. Neffke, Otto and Weyh use German social security data from 1999 to 2008 to check whether labor flows are concentrated, stable over time, and the same for different workers and for local and long-distance moves, and whether they predict regional industry growth. They find that **62%** of job switchers end up in industries holding only **5%** of employment, that this skill-relatedness network hardly changes over time and predicts local growth better than the older measures, and they publish the matrices so others can use them.

2 TWO LATER PAPERS THAT BUILD ON IT

Joyez & Laffineur (2022)
Applied Network Science 7:16

They use the same idea for occupations instead of industries, building an "Occupation Space" from the job histories of French workers between 2003 and 2015. They then show that how central an occupation is in this network is linked to its wages and to how long people in it stay unemployed.

Elekes, Tóth & Eriksson (2024)
Regional Studies 58(12), 2307–2321

They use the Neffke et al. measure but build a separate labor-flow network for each of 72 local labor markets in Sweden. They test how easily each network breaks when industries are removed, and find that regions with sturdier networks lost fewer jobs in the 2008 crisis.

3 MY NICHE: AI AND THE JOBS WORKERS CAN MOVE TO

This topic comes from my own job. At the company where I work, we are encouraged to use AI in our daily work, and teams are getting smaller, with AI expected to cover the work of the people who are no longer there. This made me wonder where the people who lose these jobs can go next. The first studies do not agree: US payroll data show fewer young workers in the most AI-exposed jobs (Brynjolfsson et al. 2025), while Danish register data show almost no effect on earnings or hours so far (Humlum & Vestergaard 2025).

RESEARCH QUESTION

For workers in occupations that are highly exposed to AI, are their most skill-related occupations also highly exposed?

HOW I WOULD STUDY IT

1. Build a skill-relatedness network between occupations from the skills each one uses, as Alabdulkareem et al. (2018) do.
2. For every occupation, take its 10 closest occupations in this network and calculate their average AI exposure.
3. Check whether occupations with high exposure have neighbors with high exposure, and whether this pattern is stronger than what random neighbors would give.

DATA AND WHO WOULD USE THE ANSWER

Data. All free to download. O*NET (US Department of Labor) lists the skills, knowledge and tasks of about 900 occupations, which is enough to build the skill network. For AI exposure, I would use the occupation scores published by Eloundou et al. (2024), and check them against the Anthropic Economic Index, which shows which O*NET tasks people actually use Claude for.

Users. Public employment services and policy makers who design retraining programs and need to know which jobs are safe to recommend, and companies like mine that are shrinking teams because of AI and need to decide where to move their staff.

RESOURCES USED

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3. Elekes, Z., Tóth, G., & Eriksson, R. (2024). Regional resilience and the network structure of inter-industry labour flows. *Regional Studies*, 58(12), 2307–2321.
4. Eloundou, T., Manning, S., Mishkin, P., & Rock, D. (2024). GPTs are GPTs: Labor market impact potential of LLMs. *Science*, 384(6702), 1306–1308.
5. Humlum, A., & Vestergaard, E. (2025). Large language models, small labor market effects. NBER Working Paper 33777.
6. Joyez, C., & Laffineur, C. (2022). The occupation space: network structure, centrality and the potential of labor mobility in the French labor market. *Applied Network Science*, 7, 16.
7. Neffke, F., Otto, A., & Weyh, A. (2017). Inter-industry labor flows. *Journal of Economic Behavior & Organization*, 142, 275–292.
8. Anthropic Economic Index (dataset). Anthropic.
9. O*NET database. U.S. Department of Labor, National Center for O*NET Development.